Exploring the interaction between compact binaries and their host galaxy

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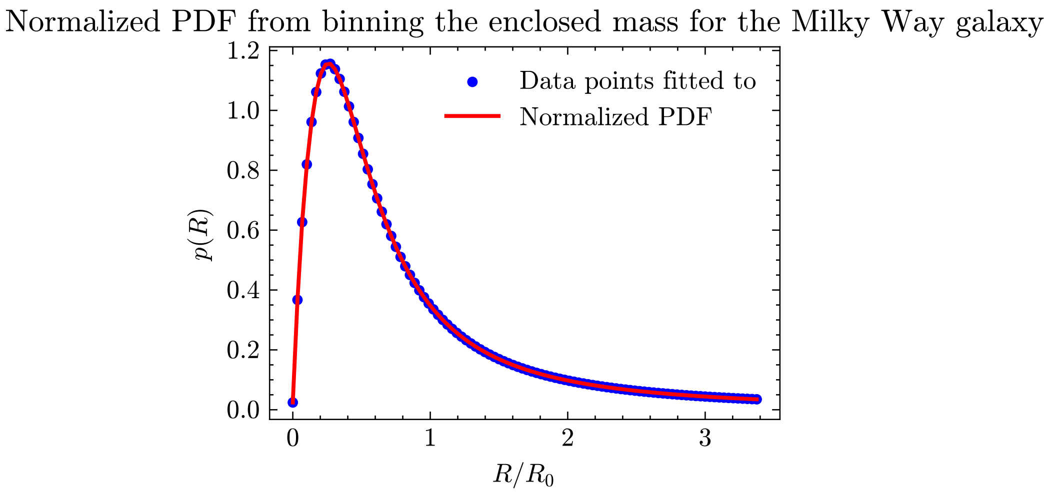
Motivation behind the project

- Infer the distribution of pulsars in the Galaxy
- Aid observations
- Compare and match simulated systems to observed pulsars
- Investigate retention fraction of different kick magnitudes
- Determine possible birth location of observed pulsars

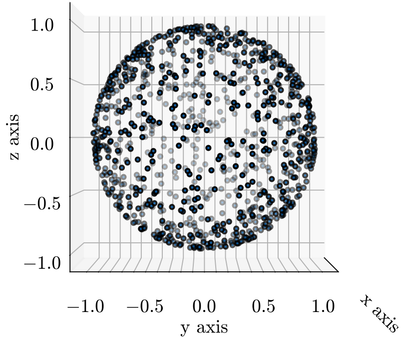
The simulation build up in previous semester



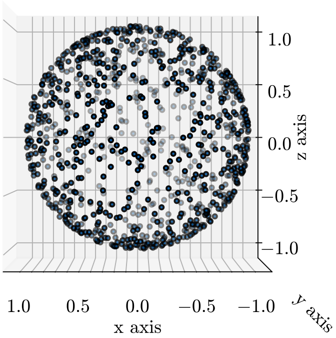
Isotoropic kick

Isotropically generated unitary directions

Top view

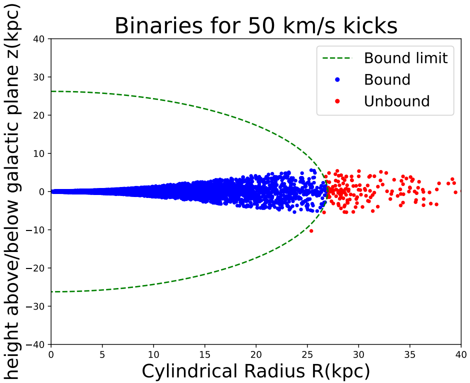


Side view



Simulated systems

- 10 000 simulated systems
- Bound limit at 27 kpc
- 90% of stellar matter
- Analytic functions
- 14 Myr time step, one point selected
- Trends outlining the potential



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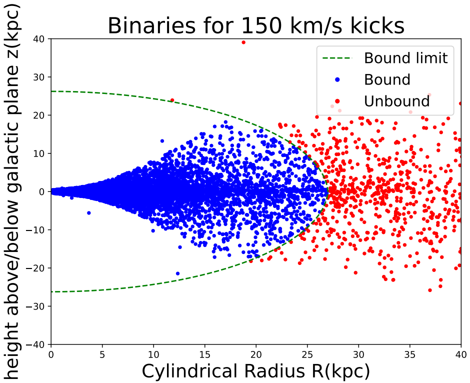

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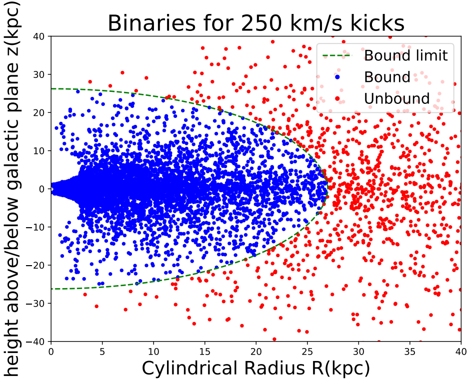
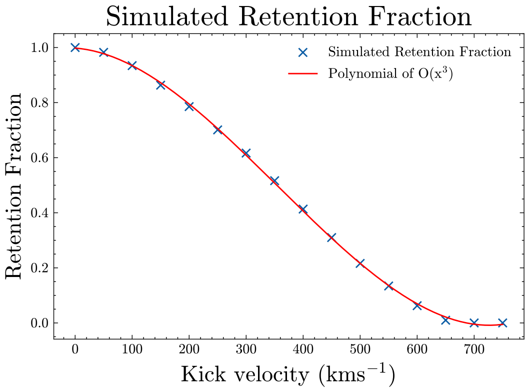


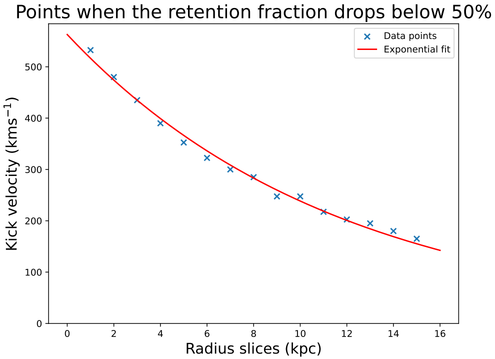
Retention fraction

- 10 000 systems at each velocity magnitude from 50 - 750 km/s in increments of 50 km/s
- Poisson error
- Follows expected trend
- Plateau at velocities above 650 km/s



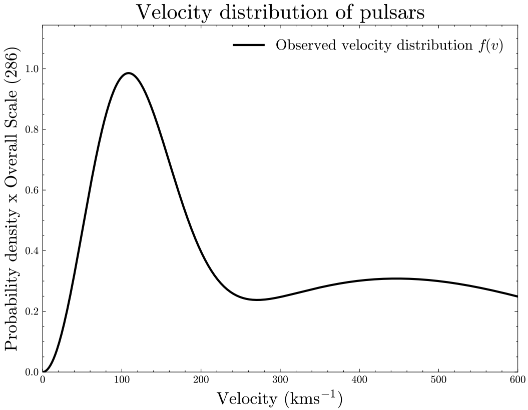
Partial retention fraction

- 16 slices at 1 kpc intervals
- From 0-1 kpc, to 14-15 kpc, with last 15+ kpc all systems
- Decreasing exponential trend, escape velocity curve



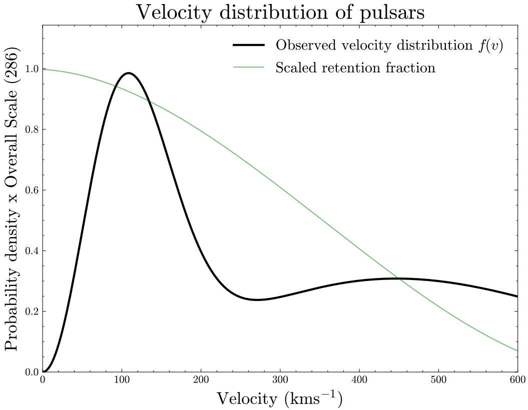
Intrinsic kick velocity distribution

- Bimodal Maxwellian with parameters: $\sigma_1 = 75\ \mathrm{kms}^{-1}$; $\sigma_2 = 316\ \mathrm{kms}^{-1}$; $f_1 = 42\ \%$; $f_2 5- = 58\ \%$
- $p\left(v, \sigma_1, \sigma_2\right) = f_1 \sqrt{\frac{2}{\pi} \frac{v^2}{\sigma_1^3}}\ e^{\left(-\frac{v^2}{2\sigma_2^2}\right)} + f_2 \sqrt{\frac{2}{\pi} \frac{v^2}{\sigma_2^3}}\ e^{\left(-\frac{v^2}{2\sigma_2^2}\right)}$



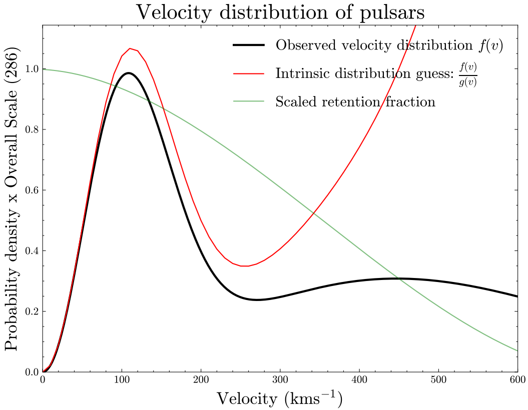
Intrinsic kick velocity distribution

- Bimodal Maxwellian
- Total retention fraction



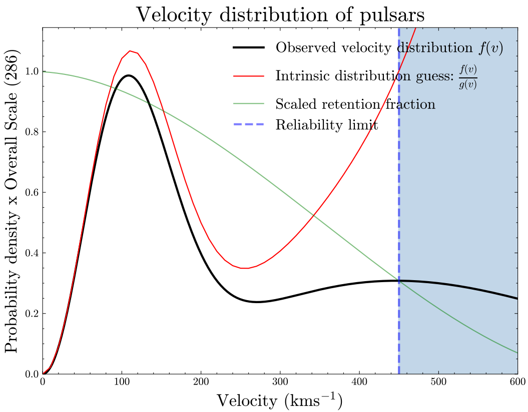
Intrinsic kick velocity distribution

- Bimodal Maxwellian
- Total retention fraction
- Intrinsic kick distribution



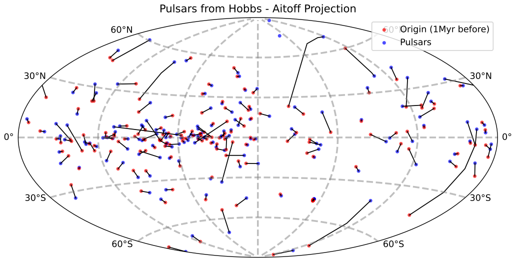
Intrinsic kick velocity distribution

- Bimodal Maxwellian
- Total retention fraction
- Intrinsic kick distribution
- Reliability limit due to retention fraction at high velocity end



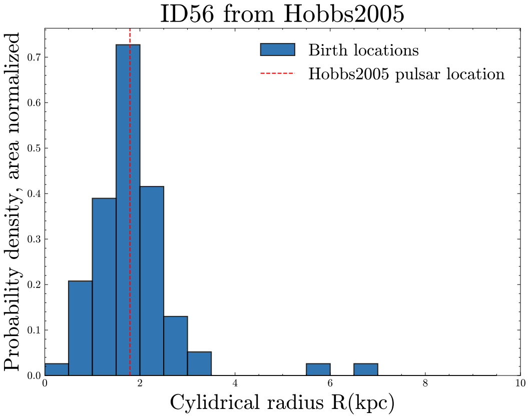
Pulsars from G. Hobbs, et al., 2005 (Hobbs 2005)

- Blue: current pulsar locations
- Red: Location 1 Myr before interpolated from proper motions



Pulsars from Hobbs2005

- Matching algorithm
- Axisymmetry of potentials, azimuth neglected
- Circle of radius 1.12 pc in R-z plane from density of simulated points
- Birth location of mathced pulsars traced back



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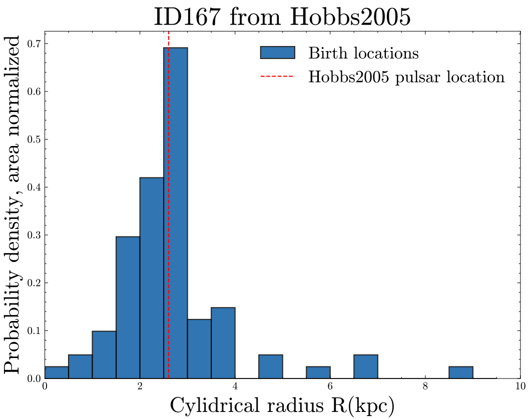

Summary

- Simulated 10 000 systems for multiple kick magnitudes
- How different kick magnitudes effect total and partial retention
- Guess of the intrinsic kick velocity distribution
- Trace back of observed pulsars
- Further velocity matching could yield interesting results